

```
● PS C:\Users\risha\Desktop\New folder> g++ .\QuickSort.cpp -o .\QuickSort.exe
PS C:\Users\risha\Desktop\New folder> .\QuickSort.exe
● =====
  QUICK SORT TIME COMPLEXITY BENCHMARK
  Array Size: 10000 elements
  =====

Running Best Case Dataset ( $O(n \log n)$ )...
Execution Time : 1.002 ms

Running Average Case Dataset ( $O(n \log n)$ )...
Execution Time : 1.022 ms

Running Worst Case Dataset ( $O(n^2)$ )...
Execution Time : 86.506 ms

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PS C:\Users\risha\Desktop\New folder> █
```

```
PS C:\Users\risha\Desktop\New folder> g++ .\BubbleSort.cpp -o .\BubbleSort.exe
PS C:\Users\risha\Desktop\New folder> .\BubbleSort.exe
=====
      BUBBLE SORT TIME COMPLEXITY BENCHMARK
      Array Size: 10000 elements
=====

Running Best Case Dataset (O(n))...
Execution Time : 0 ms

Running Average Case Dataset (O(n^2))...
Execution Time : 524.549 ms

Running Worst Case Dataset (O(n^2))...
Execution Time : 730.328 ms

Made by Rishabh Gupta
❖ PS C:\Users\risha\Desktop\New folder> █
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```
● PS C:\Users\risha\Desktop\New folder> g++ .\HeapSort.cpp -o .\HeapSort.exe
● PS C:\Users\risha\Desktop\New folder> .\HeapSort.exe
```

```
=====
HEAP SORT TIME COMPLEXITY BENCHMARK
Array Size: 10000 elements
=====
```

```
Running Best Case Dataset ( $O(n \log n)$ )...
Execution Time : 0 ms
```

```
Running Average Case Dataset ( $O(n \log n)$ )...
Execution Time : 2.999 ms
```

```
Running Worst Case Dataset ( $O(n \log n)$ )...
Execution Time : 2.02 ms
```

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```
PS C:\Users\risha\Desktop\New folder> █
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```
● PS C:\Users\risha\Desktop\New folder> g++ .\SelectionSort.cpp -o .\SelectionSort.exe
● PS C:\Users\risha\Desktop\New folder> .\SelectionSort.exe
```

```
=====
SELECTION SORT TIME COMPLEXITY BENCHMARK
Array Size: 10000 elements
=====
```

```
Running Best Case Dataset ( $O(n^2)$ )...
Execution Time : 142.005 ms
```

```
Running Average Case Dataset ( $O(n^2)$ )...
Execution Time : 156.551 ms
```

```
Running Worst Case Dataset ( $O(n^2)$ )...
Execution Time : 161.149 ms
```

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```
● PS C:\Users\risha\Desktop\New folder> g++ .\MergeSort.cpp -o .\MergeSort.exe
PS C:\Users\risha\Desktop\New folder> .\MergeSort.exe
● =====
MERGE SORT TIME COMPLEXITY BENCHMARK
Array Size: 10000 elements
=====

Running Best Case Dataset ( $O(n \log n)$ )...
Execution Time : 5.016 ms

Running Average Case Dataset ( $O(n \log n)$ )...
Execution Time : 5.998 ms

Running Worst Case Dataset ( $O(n \log n)$ )...
Execution Time : 3.998 ms

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```